

30-Day Readmission Analysis
Prepared for United Health Services (UHS)

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I. Introduction

Hospital readmissions represent one of the most costly and preventable challenges facing the modern healthcare system. Each year, millions of patients are hospitalized within 30 days of discharge, which generates billions of excess spending, and more critically, signaling failures in the care transition process that carry serious consequences for patient health and quality of life. Beyond the financial burden placed, preventable readmissions reflect unmet patient needs in the critical window between hospital discharge and recovery at home. Gaps in medication management, follow-up care, and social support can leave patients at risk long before they have fully recovered.

The Centers for Medicare & Medicaid Services (CMS) established the Hospital Readmissions Reduction Program (HERP), which financially penalizes hospitals with excess 30-day readmission rates across a set of high burden conditions, including heart failure, pneumonia, sepsis, stroke, COPD, acute myocardial infarction, and joint replacement. Under this framework, hospitals are held accountable not only for the quality of care delivered during the index stay but for the effectiveness of the transition back to the community.

United Health Services (UHS) has implemented a suite of care transition interventions to address this challenge, including the provision of medications prior to discharge, pharmacy-led transition of care (TOC) consultations, and transitional care management (TCM) follow-up visits. While each reflects best practices, their real-world effectiveness remains misunderstood within the UHS patient population. It is unclear whether these interventions meaningfully reduce the likelihood of readmission, and if so, whether their benefits are equally distributed across patients or concentrated among particular subgroups. Understanding not just whether these programs work, but whom and under what conditions, is essential for targeting resources effectively and improving outcomes across the UHS patient population. These gaps motivate the three research questions that guided this analysis.

With this data, three research questions were formulated. The first goal is to identify factors that are most significant when determining the likelihood of a patient's readmission. Along with that, our research aims to determine pre-discharge steps that will reduce readmission rate among patients with a high risk score. Lastly, evaluation of current metrics is needed. Inspection of receiving medications prior to discharge and a pharmacy consultation will be performed to see how effective they are for reducing 30-day readmissions.

II. Data Description and Preparation

The data for this analysis were provided by United Health Services (UHS) and consist of electronic health record data drawn from the Epic system. The dataset contains 5,289 adult inpatient hospitalizations that qualify as CMS anchor stays, which are index hospitalizations formally tracked for 30-day readmission under CMS readmission reduction programs. Each row represents a unique hospitalization identified by a primary index key and includes information on

admission and discharge dates, clinical diagnosis, intervention receipt, post-discharge utilization, and social determinants of health screening results.

The study population includes adult patients hospitalized for one of eight CMS-tracked major disease categories: pneumonia (PN), heart failure (HF), acute myocardial infarction (AMI), coronary artery bypass graft (CABG), stroke, sepsis, hip or knee replacement (HipKnee), and chronic obstructive pulmonary disease (COPD).

Four date variables are present in the dataset and inform both the outcome definition and the construction of time-sensitive features. The admission date and discharge date define the index hospitalization window and are used to calculate each patient's length of stay. The next ED visit date captures the first emergency department visit following anchor stay discharge and is used to characterize post-discharge utilization patterns. The TCM visit date records the first attended primary care visit after discharge performed at a UHS facility. A blank value on this variable indicated either that the patient did not attend a TCM visit or that the visit occurred outside of the UHS system. Because patients have the most control over this variable relative to the other discharge interventions, and because attendance can be influenced by factors unrelated to care quality, this variable requires careful interpretation in the regression models.

The primary outcome of interest is the 30-day readmission flag, a binary variable: 1 is the patient was readmitted to the hospital within 30 days of anchor stay discharge and 0 otherwise. A supporting variable, days between discharge and readmission, records the number of days elapsed between discharge and the next hospitalization. A null value on this variable indicated the patient was not rehospitalized following discharge and is treated as a censored observation in the main models rather than attributed as a confirmed non-event.

Three intervention variables are central to this analysis. The medications prior to discharge flag is a binary indicator: 1 if the patient received prescribed medications before leaving the hospital and 0 otherwise. The TOC consultation flag indicates whether the patient received a pharmacy-led consultation on their medication protocol prior to discharge. This variable serves as the primary interaction term of interest across the regression models. The TCM visit date is used to construct binary time-windowed indicators reflecting whether a follow-up visit occurred within 7, 10, or 14 days of discharge, described further in the feature engineering section. The clinician type performing the TCM visit is also recorded, distinguishing between physicians (MD or DO) and advanced practice providers (NPs and PAs).

Each hospitalization is assigned a major disease category reflecting the primary reason for admission, drawn from the eight CMS-tracked conditions listed above. Epic's readmission risk score provides a continuous predicted probability of readmission on a scale of 0 to 100, generated at the time of the index stay. A corresponding categorical variable classifies patients as low, medium, or high risk based on Epic's internal thresholds.

Demographic variables include age group, categorized as 18–39, 40–49, 50–59, 60–69, 70–79, and 80+; biological sex; self-identified race, categorized as white or non-white for analysis; primary language, distinguished as English or non-English; marital status; self-identified sexual orientation; and smoking status recorded at admission.

Thirteen binary social determinants of health flags are included in the dataset, each reflecting whether a patient was screened as medium or high risk on a given social domain at the time of their index stay. The domains captured are alcohol use, depression, financial resource strain, food insecurity, health literacy, housing stability, intimate partner violence, physical activity, social connections, stress, tobacco use, transportation needs, and utilities. Each flag is coded 1 for medium or high risk and 0 for low risk. These variables are incorporated into the analysis in two ways: individually as covariates in the regression models to control for social risk, and collectively to construct a composite SDOH burden score and clustering variable that groups patients by their overall social risk profile, described further in the feature engineering section.

Before analysis, the dataset was examined for inconsistencies and missing values across all variables. An initial check revealed a small number of errors in the 30-day readmission flag, specifically in the cases where the flag was coded 0 despite the days between discharge and readmission value, indicating that a readmission had occurred within 30 days. These records were corrected by overwriting the flag to 1 using the days between discharge and readmission variable as the ground truth. A second check identified cases in which the readmission flag was coded 1, but the days between discharge and readmission value were null, indicating a data inconsistency.

Missing values were assessed across all columns. Variables with low rates of missingness, such as the readmission risk value and readmission risk category, were inspected at the individual level to determine whether missingness was random or patterned. The days between discharge and readmission variable were not treated as missing in the traditional sense. A null value on this variable indicates the patient was not rehospitalized following the discharge and is therefore censored at 30 days in the survival time construction rather than imputed or excluded.

All date columns were converted to datetime format using pandas to_datetime with errors coerced, ensuring that non-parseable values were handled. The primary date transformation involved computing the number of days between the discharge date and the TCM visit date, stored as a derived variable called DaysToTCM. This served as the basis for the TCM time window indicators that were constructed in the feature engineering step. A separate check confirmed that no records constrained a discharge date occurring before the admission date, which verifies the internal consistency of the hospital date range.

Three binary indicators were constructed to capture whether a patient received a TCM follow-up visit within the clinically relevant windows after discharge. A patient was coded 1 on TCM_7d if their DaysToTCM value fell between 0 and 7 days inclusively, and 0 otherwise. The same logic was applied to construct TCM_10d and TCM_14d using 10-day and 14-day thresholds, respectively. Patients with no recorded TCM visit received a value of 0 across all three indicators. TCM_7d serves as the primary specification, with TCM_10d and TCM_14d used for sensitivity testing of the time window assumption.

A composite transition bundle score was constructed by summing the three binary intervention indicators: medications prior to discharge, TOC consultation, and TCM within the given time window. This produces an integer score ranging from 0 to 3, where 0 indicates no interventions received and 3 indicates all three interventions received. Three versions of the bundle score were created corresponding to each TCM time window: TOC_bundle_7d, TOC_bundle_10d, and TOC_bundle_14d. This variable is used in the dose-response analysis to test whether receiving a greater number of interventions is associated with progressively lower readmission odds.

The thirteen binary SDOH flags were used to construct two complementary social risk variables. A composite SDOH burden score was created by summing up the number of domains on which a patient was flagged as medium or high risk, producing an integer score ranging from 0 to 13. This score captures overall social vulnerability and is used as a continuous covariate in the regression models. The thirteen SDOH flags were also grouped into four thematic clusters based on their conceptual domain: Behavior (alcohol use, physical activity, tobacco use), Mental (depression, stress, social connections, intimate partner violence), Financial (financial resource strain, utilities, food insecurity, housing stability), and Access (health literacy, transportation needs). Patients were assigned to a cluster based on their dominant area of social risk. This clustering is used in subgroup-level analysis and moderation models to examine whether intervention effectiveness differs across social risk profiles.

The age group variable was encoded as an ordered numeric factor. The six categories were mapped as follows:

- 18-39 = 1
- 40-49 = 2
- 50-59 = 3
- 60-69 = 4
- 70-79 = 5
- 80+ = 6

This ordering preserves the monotonic relationship between age and readmission risk and allows the variable to be interpreted directionally in the regression models.

The MDC category variable was transformed using one-hot encoding, producing eight binary indicator columns, one for each CMS-tracked disease category. This approach allows each disease category to enter the regression model independently with its own coefficient, since the disease category has no natural ordering. The resulting columns are MDC_AMI, MDC_CABG, MDC_COPD, MDC_HF, MDC_HipKnee, MDC_PN, MDC_Sepsis, and MDC_Stroke.

III. Methodology

Prior to modeling, a descriptive analysis was conducted to characterize the study population and establish baseline readmission patterns across key subgroups. Overall, 30-day readmission rates were calculated for the full sample, followed by breakdowns across the four

dimensions: MDC category, readmission risk category, age group, and SDOH cluster. For each subgroup, both the proportion of the overall sample represented by that group and the within-group readmission rate were reported. This allows for a distinction between groups that contribute most to overall readmission volume and groups that carry the highest individual readmission risk.

Treatment prevalence was examined by calculating the proportion of patients who received each intervention: medications prior to discharge, TOC consultation, and TCM within 7, 10, and 14 days of discharge. This provides context for interpreting the regression results and highlights differences in intervention delivery across the sample/

Finally, a comparison of patient characteristics was conducted across treated and untreated groups for each intervention. This comparison focused on two key dimensions: the Epic readmission risk score and the composite SDOH burden score. Differences in these characteristics between treated and untreated patients indicate the degree to which intervention receipt was not randomly distributed across the sample, motivating the balance diagnostics described below.

To assess the degree of systematic difference between treated and untreated patients, standardized mean differences (SMDs) were calculated for two key covariates: the Epic readmission risk score and the composite SDOH burden score. The SMD expresses the difference in means between two groups as a fraction of the pooled standard deviation, making it comparable across variables measured on different scales. An SMD of 0 indicates no difference between groups, while values below 0.1 are generally considered indicative of adequate covariate balance. These two variables were selected for balance diagnostics because they represent the dimensions within which treated and untreated patients are more likely to differ systematically: clinically through risk score and socially through SDOH score.

SMDs were computed both before and after applying inverse probability weights to evaluate whether weighting improved covariate balance between treated and untreated patients. Before weighting, the readmission risk score showed an SMD of 0.15, and the SDOH burden score showed an SMD of 0.20, indicating moderate imbalance between treated and untreated patients on both dimensions. After weighting, the SMD for the risk score was reduced to 0.05 and 0.004 for the SDOH burden score, which both fall within the 0.1 threshold. This reduction in imbalance across both variables indicates that the weighting procedure achieved adequate covariate balance and supports a more credible comparison between treated and untreated patients in the regression models.

The primary analysis uses logistic regression to estimate the association between each care transition intervention and the odds of 30-day readmission. The model is as follows:

$$\begin{aligned} \text{logit}(P(\text{Readmit30} = 1)) = & \beta_0 + \beta_1(\text{MedsPriorToDC}) + \beta_2(\text{TOCConsult}) \\ & + \beta_3(\text{TCM7d}) + \beta_4(\text{SVIz}) + \beta_5(\text{AgeGroup}) + \beta_6(\text{SexMale}) + \beta_7(\text{RaceNonWhite}) \\ & + \beta_8(\text{MDCCategory}) \end{aligned}$$

The three intervention variables, medications prior to discharge, TOC consultation, and TCM within 7 days, are the binary predictors of the model. A z-scored social vulnerability index (SVI_z) is included to control for community-level social risk. Age group is an ordered numeric variable as described in the feature engineering section. Sex and race categories were binary encoded, with Sex_Male coded 1 for male patients and Race_NonWhite coded 1 for non-white patients. MDC category enters as a set of binary indicator variables with Sepsis serving as the reference category. The model was estimated using HC1 robust standard errors to account for heteroskedasticity and within-subject correlation.

The covariate set was constructed to isolate the association between intervention receipt and readmission from confounding driven by patient-level differences in risk and social circumstances. The Epic readmission risk score was excluded from the final model because it is itself a function of the outcome we are trying to predict. Including it would introduce circularity and inflate the explanatory power of the other predictors. Rather than including the 13 individual SDOH flags or a raw composite burden score, social risk is controlled for through SVI_z, a standardized social vulnerability index score. SVI_z captures the same underlying social risk information as the SDOH flags in a single standardized variable, avoiding the multicollinearity and dimensionality issues that arise from including many correlated binary indicators in the same model. Age group, sex, and race category are included to account for known demographic differences in healthcare utilization and readmission outcomes. MDC category is included because the primary diagnosis is one of the strongest determinants of baseline readmission risk, and differences in the mix of conditions across treated and untreated groups could otherwise confound the intervention estimates.

Results are reported as both odds ratios and average marginal effects. Odds ratios convey the relative change in the odds of readmission associated with each predictor. Average marginal effects translate these estimates into average percentage point changes in the probability of readmission, which are more directly interpretable for clinical and policy audiences.

The moderation analysis uses a K-Modes clustering algorithm applied to a subset of six SDOH variables, depression, financial resource strain, food insecurity, health literacy, housing stability, transportation needs, and utilities, to group patients into distinct social risk profiles. K-Modes was selected over K-Means because the SDOH variables are binary rather than continuous. Three clusters were selected because they produced the cleanest separation between patient groups. The resulting cluster assignments were one-hot encoded and entered into a separate logistic regression model with 30-day readmission as the outcome, with cluster 0 serving as the reference group.

The three clusters identified were labeled based on their profiles:

- Cluster 0: General Population, characterized by low prevalence across all SDOH domains

- Cluster 1: Economic Hardship, characterized by high prevalence of financial resource strain, housing instability, food insecurity, transportation needs, and utilities strain
- Cluster 2: Depression-Dominant, characterized by universally high depression flags with relatively lower prevalence across the financial and access domains

The clustering model tests whether membership in a distinct social risk profile is independently associated with readmission odds relative to the general population baseline. A significant positive coefficient on a cluster indicator indicates that patients in that social risk group have higher odds of readmission than general population patients after accounting for cluster membership. Results are reported as odds ratios and percentage increase in odds relative to the reference cluster to aid interpretation.

IV. Findings

The data has a total of 5,289 patients and 14.3% of these patients were readmitted with its 95% confidence interval falling between 13.4% and 15.3% . Below, we analyzed the readmission patterns across certain patient diagnostics and qualities.

The table below displays the readmission rate and percent of readmissions by the eight MDC categories given the dataset, organized by readmission rate from highest to lowest percentage. Two categories have over a 20% readmission rate (COPD and Heart Failure). Additionally the top four MDC Categories of readmission rate are also in the top percent of all readmissions.

Table 1: Readmission Results by MDC Category

MDC Category	Readmission Rate	% of All Readmissions
COPD	24.5%	13.2%
Heart Failure (HF)	21.0%	23.6%
Pneumonia (PN)	19.8%	15.1%
Sepsis	14.8%	27.7%
Hip/Knee	12.3%	1.8%
CABG	10.3%	3.1%
AMI	7.7%	5.9%
Stroke	8.8%	9.6%

Next, evaluation was performed on each age group. As age increases, the percentage of all readmission and percent admitted also increases. This is to be expected, as most MDC cases with high readmission rates occur in older individuals. The steepest jump occurs between the 50-59 and 60-69 age bands (3.7% difference), suggesting that patients crossing into that age range represent a key intervention threshold.

Table 2: Readmission Rate by Age Group

Age Group	Readmission Rate	% of All Readmissions
80+	16.4%	33.2%
70-79	16.3%	29.2%
60-69	15.9%	23.7%
50-59	12.2%	8.0%
40-49	10.5%	3.9%
18-39	7.3%	2.0%

The existing risk scoring system represents discrimination between patients. High risk patients have a readmission rate of 19.0% compared to 16.4% and 9.7% for medium risk and low risk respectively. However, seeing that patients classified as high risk represent 43.8% of readmissions, this shows that a substantial portion of readmitted patients are initially classified as medium or low risk. Therefore, this reinforces the values of SDOH categories beyond risk score.

For categorization based on SDOH burden, transportation barriers, food insecurity, and housing instability are associated with high readmission rates. These are the factors we determined are actionable pre-discharge compared to clinical diagnoses alone.

Table 3: Readmission Rate by SDOH Factor

SDOH Factor	% of Patients with Flag	Readmission Rate Among Flagged Patients
Alcohol Use	10.74%	14.63%
Depression	17.85%	18.50%
Financial Strain	29.09%	22.49%
Food Insecurity	17.98%	24.74%
Health Literacy	7.62%	20.13%
Housing Instability	19.10%	23.72%

SDOH Factor	% of Patients with Flag	Readmission Rate Among Flagged Patients
Intimate Partner Violence	1.75%	18.42%
Low Physical Activity	88.51%	18.37%
Social Isolation	73.16%	19.78%
Stress/Anxiety	42.57%	24.06%
Tobacco Use	70.66%	16.04%
Transportation Barriers	25.84%	25.59%
Financial Utility Strain	8.11%	25.10%

Across the 5,289 patients that were offered the interventions of Meds Prior to Discharge and TOC Consultation interventions, the following trends were observed during initial analysis. For the MedsPriorToDC intervention variable, 24.09% of patients received medication and the other patients did not. With this being considered, 15.19% of the patients that received medication were readmitted, and 8.57% of those that did not receive medication prior to discharge were readmitted. This suggests that patients receiving their medication may have represented a population with higher risk and more complex cases, contributing to the increased rate.

As per the TOC Consultation, 37.14% had a consultation and the other patients did not have a consultation. Out of the patients that had a consultation before discharge, 11.95% were readmitted. For the other patients that did not receive one, 16.55% were readmitted. By observation, these results say that TOC consultations may be associated with reduced readmissions. This is mainly due to improved discharge planning and follow-up care by UHS.

Table 4: Balance Diagnostics

Variable	Treated Patients Mean	Untreated Patients Mean	SMD (pre-weighting)	SMD (post-weighting)
Risk Score	17.93	19.64	0.15	0.05
SDOH Score	3.18	3.67	0.20	0.004

The balance diagnostics table shows the standardized mean differences (SMDs) for the observed characteristics between treated and untreated patients in the risk and SDOH score categories. Prior to weighting, treated patients had a lower average risk score and SDOH Score compared to untreated patients, indicating that there is meaningful difference between these two groups. After weighting, balance between the groups improved drastically. The Risk Score SMD decreased from 0.15 to 0.05, while the SDOH Score SMD decreased from 0.20 to 0.004. With

the post-weighting SMD values falling below 0.10, this is considered acceptable. We can say this weighting process was effective in removing observable differences in the above patient groups.

As a result, comparisons of readmission outcomes are less likely to be driven by their differences in baseline patient characteristic and may provide a more accurate estimation of how the intervention affects readmission risk and rate.

Table 5: Logistic Regression Results

Index	Coefficient	Odds Ratio	AME	Confidence Interval (Lower)	Confidence Interval (Higher)	P-Value
const	-3.318	0.036	-	0.010	0.136	< 0.001
AnchorStayMedsPriorTo DC	0.871	2.389	0.105	0.670	8.513	0.179
AnchorStayTOCConsult	-0.123	0.8845	-0.015	0.722	1.083	0.235
TCM_7d	-0.786	0.456	-0.095	0.334	0.622	< 0.001
SVI_z	0.457	1.579	0.055	1.469	1.698	< 0.001
AgeGroup	0.165	1.179	0.020	1.106	1.257	< 0.001
Sex_Male	0.052	1.054	0.006	0.900	1.235	0.516
Race_NonWhite	-0.338	0.713	-0.041	0.505	1.007	0.055
MDC_AMI	-0.606	0.545	-0.073	0.386	0.770	< 0.001
MDC_CABG	-0.318	0.727	-0.038	0.467	1.1316	0.158
MDC_COPD	0.374	1.453	0.045	1.103	1.916	0.008
MDC_HF	0.289	1.335	0.035	1.0648	1.674	0.012
MDC_HipKnee	-0.393	0.675	-0.047	0.3734	1.221	0.194
MDC_PN	0.197	1.218	0.024	0.943	1.574	0.132
MDC_Stroke	-0.450	0.638	-0.54	0.470	0.866	0.004

Among the intervention variables, TCM within 7 days was significantly associated with reduced odds of readmission. Its corresponding AME suggests that having a TCM follow-up within seven days is associated with an estimated 7.8% reduction in the probability of readmission while other variables stay constant. TOC consultation is not statistically significant with a high p-value, noting that there is no measurable association with readmission after

adjustments. Similarly, MedsPriorToDC has a high p-value showing no statistical significance, even though an odds ratio greater than 1 may suggest that patients that receive this intervention have a higher baseline risk.

Through the MDCs, patients with acute myocardial infarction (AMI) have significantly lower odds of readmission compared to the reference category. Other categories, including COPD and HF, approach statistical significance but do not meet the 0.05 threshold. Overall, basic regression findings suggest that timely TCM follow-up is the strongest intervention when pertaining to reducing readmission risk.

The multivariable logistic model demonstrates an acceptable overall fit for predicting 30-day hospital readmission. It produced a psuedo- R^2 of 0.0697. This signifies a moderate performance. In other words, the included interventions, risk measures, and demographics explain a meaningful proportion of the variation in readmission outcomes.

Several interventions and variables did not achieve the significance threshold after adjustment for characteristics. Precisely, MedsPriorToDC and TOCConsult were not significantly associated with readmission risk. Although these findings are not significant, they should not be interpreted as ineffective. Instead, the results from our research indicate that a reliable independent associate was not detected after controlling for other covariate factors. For example, TOCConsult has an odds ratio below 1, noting a protective association even with a confidence interval including 1.0. This shows that there is uncertainty regarding the actual effect of the intervention.

Overall, when evaluating all the null findings in this model, the following should be considered: effect direction, confidence intervals, and clinical plausibility instead of relying on the significance thresholds.

Sensitivity analysis was performed on each TCM follow-up period in order to determine the robustness of our primary findings. Across all of these models, TCM remained directionally associated with reduced odds of 30-day readmission, supporting the use of this intervention across various time windows. The strongest protective association was seen in the seven-day time period. This indicates that earlier engagement post-discharge can provide the greatest benefit when the goal is to reduce the risk of readmission. Even though we see the effect diluted in 10-day and 14-day models, the pattern stayed consistent. Overall, these analyses strengthen the confidence of the TCM finding and hope that continuous use of this intervention can cause readmission rates to decline in the future.

In addition, the TOC pharmacy consult (AnchorStayTOCConsult) showed no significant independent effect on readmission rate. However, these results cannot be generalized as ineffective for two reasons. First, 75.9% of readmitted patients did not receive a consultation. With such a small treated group, these results cannot be definite. Furthermore, the TOC consultation and economic hardship interaction term are approaching significance in our model. This suggests that having a consultation may disproportionately benefit the patients with the most needs, an area that will be further discussed later. An expansion of TOC Consult coverage is needed across SDOH-burdened patients before any final conclusions are made.

The AnchorStayMedsPriorToDC is a binary variable in this dataset. Only 35 patients (0.66%) did not receive medications prior to discharge. It is this imbalance that causes this variable to not function as a meaningful independent predictor in our regression model. The coefficient is directionally positive with an odds ratio of 2.39. These numbers come from a small control group rather than a true effect. It should not be used as an intervention metric without increased variation in the data.

Moderation analysis was conducted to determine whether TCM was effective across different subgroups of patients. Significant interactions were identified for age and baseline readmission risk value. This shows that TCM effectiveness is not uniform over all patient populations.

The interaction between TCM and patients over the age of 70 is statistically significant, with an odds ratio of 1.87 and a p-value of 0.017. Specifically, the intervention was associated with an approximate 71% reduction in readmission for patients younger than 70 compared to the 34% reduction among patients 70 and over. Even though TCM is noted as beneficial for elderly patients, it appears substantially more effective for younger populations. The reason for this is unknown, but we can infer that it may be due to greater functional independence or a lower comorbidity burden.

The second significant interaction is found between TCM and readmission risk value. For this category specifically, TCM shows different effectiveness across the given risk categories. Patients categorized as high risk that receive transitional care management show considerably reduced readmission rates when compared to the high-risk patients who do not.

V. Discussion

The most actionable finding from this study was the strong association between timely transitional care management, usually within seven days, and reduced 30-day hospital readmissions. Our findings suggest that early post-discharge follow-ups represent one of the most important resource allocation strategies for cutting preventable readmissions for UHS patients.

Across the three interventions studied (TCM, TOCConsult, and MedsPriorToDC), TCM has provided the strongest evidence of being impactful on patients. This means that a targeted follow-up is an important intervention measure. On the other hand, the lack of statistical significance for TOCConsult and MedsPriorToDC should not allow us to think these interventions are not useful. Alternatively, they have been disproportionately used in clinically complex scenarios with a higher readmission risk. The overlap between pathways can reduce the ability to separate the effect of each program individually.

In addition, there are some subgroups that may require more interventions than others and vice versa. To begin, high-risk patients demonstrate the greatest reduction in readmission risk when given a TCM intervention. This claim supports the use of risk-ranked transitional care models. Additionally, patients over the age of 70 may require more intensive transitional care

management procedures due to caregiver dependence, functional limitation, or other outside factors not denoted in the data.

This analysis has some limitations that should be considered when interpreting findings and implementing recommendations. All analyses are observational. Despite propensity score weighting to balance treated and untreated groups on observed confounders (risk score, SDOH score), unmeasured confounding may persist. The significant SMD reduction post-weighting (Risk: 0.15 to 0.05; SDOH: 0.20 to 0.004) provides confidence but does not eliminate the possibility of residual bias. Furthermore, TCM visit date is missing for 78.7% of records. Patients without a TCM date were treated as not receiving a TCM visit ($TCM_7d = 0$). If some of these patients did receive a visit that was not recorded, the estimated effect of TCM is downwardly biased - the true effect may be even larger than reported.

With only 35 patients (0.66%) not receiving medications before discharge, this variable cannot be meaningfully analyzed as an intervention. Any reported odds ratio for this variable should be disregarded for practical decision-making. 1,033 records had apparent discharge date parsing issues. While these were resolved through standardized datetime conversion, residual data quality concerns in the source system may affect survival time calculations or TCM window assignments for a subset of records.

We report odds ratios and average marginal effects, not causal treatment effects. TCM may be systematically scheduled for patients who are already on a better recovery trajectory (healthy worker bias). A randomized or quasi-experimental design (e.g., instrumental variable) would be required to establish true causal effects.